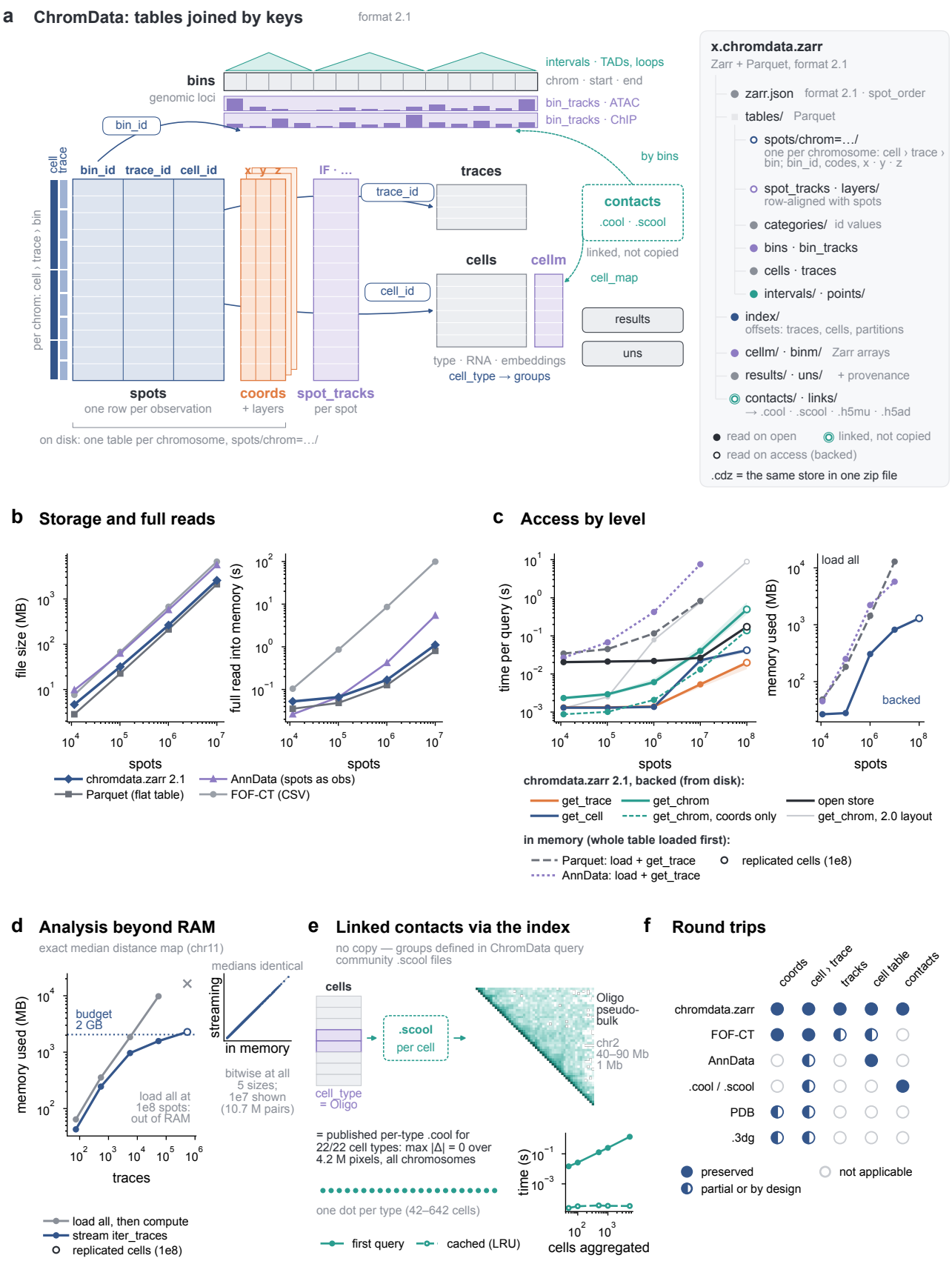

a ChromData: tables joined by keys
format 2.1
bins
genomic loci
intervals · TADs, loops
chrom · start · end
bin_tracks · ATAC
bin_tracks · ChIP
bin_id
by bins
cell
trace
per chrom: cell › trace › bin
bin_id trace_id cell_id
x y z
IF · …
traces
trace_id
contacts
.cool · .scool
linked, not copied
cells
cellm
cell_id
cell_map
results
uns
type · RNA · embeddings
cell_type → groups
spots
one row per observation
coords
+ layers
spot_tracks
per spot
on disk: one table per chromosome, spots/chrom=…/
x.chromdata.zarr
Zarr + Parquet, format 2.1
zarr.json format 2.1 · spot_order
tables/ Parquet
spots/chrom=…/
one per chromosome: cell › trace › bin; bin_id, codes, x · y · z
spot_tracks · layers/
row-aligned with spots
categories/ id values
bins · bin_tracks
cells · traces
intervals/ · points/
index/
offsets: traces, cells, partitions
cellm/ · binm/ Zarr arrays
results/ · uns/ + provenance
contacts/ · links/
→ .cool · .scool · .h5mu · .h5ad
read on open
linked, not copied
read on access (backed)
.cdz = the same store in one zip file
b Storage and full reads
file size (MB)
full read into memory (s)
spots
chromdata.zarr 2.1
AnnData (spots as obs)
Parquet (flat table)
FOF-CT (CSV)
c Access by level
time per query (s)
memory used (MB)
load all
backed
spots
chromdata.zarr 2.1, backed (from disk):
get_trace
get_chrom
open store
get_cell
get_chrom, coords only
get_chrom, 2.0 layout
in memory (whole table loaded first):
Parquet: load + get_trace
replicated cells (1e8)
AnnData: load + get_trace
d Analysis beyond RAM
exact median distance map (chr11)
budget 2 GB
load all at 1e8 spots: out of RAM
memory used (MB)
traces
medians identical
streaming
in memory
bitwise at all 5 sizes; 1e7 shown (10.7 M pairs)
load all, then compute
stream iter_traces
replicated cells (1e8)
e Linked contacts via the index
no copy — groups defined in ChromData query
community .scool files
cells
.scool
per cell
cell_type = Oligo
Oligo pseudo-bulk
chr2 40–90 Mb 1 Mb
= published per-type .cool for 22/22 cell types: max |Δ| = 0 over 4.2 M pixels, all chromosomes
one dot per type (42–642 cells)
time (s)
cells aggregated
first query
cached (LRU)
f Round trips
coords
cell › trace
tracks
cell table
contacts
chromdata.zarr
FOF-CT
AnnData
.cool / .scool
PDB
.3dg
preserved
not applicable
partial or by design